

Complex Numbers

Recall Notation & Definitions:

$$(1) \quad z \in \mathbb{C} \Rightarrow z = |z| \cdot e^{i \arg(z)}.$$

$$|z| \in \mathbb{R}, \quad |z| \geq 0$$

$$\arg(z) = \left\{ \underbrace{\text{Arg}(z)}_{\downarrow} + 2\pi k : k \in \mathbb{Z} \right\}.$$

Principal phase, $\text{Arg}(z) \in [-\pi, \pi)$

(2) Log : denotes the real logarithm.

(3) Complex logarithm: $\log$

$$\log(z) = \left\{ \underbrace{\text{Log}|z| + i \text{Arg}(z)}_{\text{Principal logarithm}} + i 2\pi k : k \in \mathbb{Z} \right\}$$

Principal logarithm
Denote this as $\text{Log}(z)$

(4) Properties:

$$* \quad \log(z_1 z_2) = \log(z_1) + \log(z_2)$$

= set summation

$$\text{Aside: } A+B = \{a+b : a \in A, b \in B\}$$

$$* \quad \log(z_1/z_2) = \log(z_1) - \log(z_2)$$

$$* \quad \text{Warning: In general, } \log(z^p) \neq p \log(z).$$

But: if $\frac{1}{p}$ is an integer:

$$\log(z^p) = p \log(z).$$

$$* e^{\log z} = z$$

$$(5) e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

$$e^{z_1}, e^{z_2} = e^{z_1 + z_2}.$$

$$(6) \text{ For } z, c \in \mathbb{C}:$$

$$z^c = e^{c \log z} \quad (\text{this is the definition})$$

I. Examples for z^c

$$z \rightarrow \log(z) = \{ \text{Log}(z) + i2\pi k : k \in \mathbb{Z} \}$$

$$c \log(z) = \{ c \text{Log}(z) + i2\pi kc : k \in \mathbb{Z} \}$$

$$z^c = e^{c \log z} = \{ e^{c \text{Log}(z)}, e^{i2\pi kc} : k \in \mathbb{Z} \}.$$

(i) What if c is an integer?

$$\text{If } c \text{ is an integer then } z^c = e^{c \log(z)}$$

This is exactly one value (i.e. this is not multivalued).

In fact, the only values of c for which z^c is single-valued are $c \in \mathbb{Z}$.

When $c \in \mathbb{Z}$:

$$z^c = \begin{cases} z \times z \times \dots \times z, & c > 0 \\ z^{-1} \times z^{-1} \times \dots \times z^{-1}, & c < 0 \end{cases}$$

② Suppose $c = \frac{1}{n}$, where $n > 0$ and $n \in \mathbb{Z}$.

$$z^c = z^{1/n} = \left\{ e^{\frac{1}{n} \log z} \cdot e^{i2\pi k/n}; k \in \mathbb{Z} \right\}.$$

Here, $z^{1/n}$ is a n -valued function

$$z^{1/n} = \left\{ e^{\frac{1}{n} \log z} \cdot e^{i2\pi k/n}; k=0, \dots, n-1 \right\}.$$



These n numbers are precisely the n distinct solutions to $x^n - z = 0$.

③ $z^0 = ?$

$$\begin{aligned} z^0 &= e^{0 \log z} = \left\{ e^{0 \log z} \cdot e^{i2\pi k \cdot 0}; k \in \mathbb{Z} \right\} \\ &= 1 \end{aligned}$$

Properties of z^c :

1. In general $z^a \cdot z^b \neq z^{a+b}$.

$$z^a \cdot z^{-b} \neq z^{a-b}.$$

2. $z^{-a} = \frac{1}{z^a}$

3. $(z_1 \cdot z_2)^a = z_1^a \cdot z_2^a$

4. $(z_1/z_2)^a = z_1^a \cdot z_2^{-a}$.

5. In general, $(z^a)^b \neq z^{ab}$.

consider the example where $a = n$, $n \in \mathbb{Z}$, $n > 0$

$$\text{ \& b = } 1/n$$

$$z^{ab} = z^1 \rightarrow \text{one single value} = z$$

$$(z^a)^b = (z^n)^{1/n}$$

$$= (z \times \dots \times z)^{1/n}$$

$\rightarrow$ collection consisting of n distinct values

$$z^{ab} = \text{one single number.}$$

Summary

1. $z^a = e^{a \log z}$

2. $(z_1 z_2)^a = z_1^a \cdot z_2^a$

3. $(z_1 / z_2)^a = z_1^a \cdot z_2^{-a}$

4. $(1/z)^a = z^{-a}$

5. $(z^a)^b \neq z^{ab}$ in general

But, equality works if $a = \pm 1$ or if $b \in \mathbb{Z}$

6. $z^a \cdot z^b \neq z^{a+b}$ in general

$z^a \cdot z^{-b} \neq z^{a-b}$ in general.

II. Fourier Series.

Recall: x is a T -periodic function.

The n^{th} Fourier series coefficient of x

is:

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega t} dt, \quad \omega = \frac{2\pi}{T}$$

$$= \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt.$$

$$x \xleftrightarrow{\text{F.S.}} C_n$$

$$x' \xleftrightarrow{\text{F.S.}} \frac{dn}{j\omega}$$

Fourier series of $\frac{dx}{dt}$:

Assuming that $\frac{dx}{dt}$ exists at all $t \in \mathbb{R}$.

Denote this function as $x'(t)$.

$$\text{let } x' \xleftrightarrow{\text{F.S.}} d_n \quad \left\{ \begin{array}{l} C_n = \frac{dn}{j\omega} \end{array} \right.$$

For $n=0$:

$$d_0 = \frac{1}{T} \int_0^T x'(t) e^{-j0\omega t} dt$$

$$= \frac{1}{T} \int_0^T x'(t) \cdot e^{-j0\omega t} dt, \quad n=0$$

$$= \frac{1}{T} \int_0^T x'(t) dt = \frac{1}{T} \int_0^T d(x(t))$$

$$= \frac{1}{T} [x(t)]_0^T = 0$$

For $n \neq 0$:

$$C_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt$$

$$Tc_n = \int_0^T x(t) \cdot d \left[\frac{e^{-jn\omega t}}{-jn\omega} \right]. \quad \left. \begin{array}{l} \text{under} \\ = uv - \int v du \end{array} \right\}$$

$\xleftrightarrow{u} \quad \xleftrightarrow{v}$

$$= \left[x(t) \cdot \frac{e^{-jn\omega t}}{-jn\omega} \right]_0^T + \frac{1}{jn\omega} \int_0^T e^{-jn\omega t} \cdot x'(t) dt$$

$$= \frac{1}{jn\omega} \int_0^T x'(t) \cdot e^{-jn\omega t} dt$$

$$Tc_n = \frac{1}{jn\omega} \cdot T d_n$$

$$\therefore d_n = jn\omega \cdot c_n = jn \cdot \frac{2\pi}{T} \cdot c_n$$

$$\Rightarrow d_n = j \frac{n \cdot 2\pi}{T} \cdot c_n \text{ for all } n \in \mathbb{Z}$$

Corollary: If x'' exists at all t and

$$x'' \xleftrightarrow{\text{F.S.}} b_n$$

$$b_n = j \cdot \frac{n \cdot 2\pi}{T} \cdot d_n$$

$$= \left(j \frac{n \cdot 2\pi}{T} \right)^2 \cdot c_n$$

Corollary: $x \xleftrightarrow{\text{F.S.}} c_n$

$$x' \xleftrightarrow{\text{F.S.}} j \frac{2\pi n}{T} \cdot c_n$$

Now suppose $c_0 \neq 0$

$$\|x'\|^2 = T \sum_{n=-\infty}^{+\infty} |d_n|^2$$

$$= T \sum_{n \neq 0} \left| j \frac{2\pi n}{T} c_n \right|^2$$

$$= T \sum_{n \neq 0} \frac{4\pi^2 n^2}{T^2} |c_n|^2$$

$$\geq T \sum_{n \neq 0} \frac{4\pi^2}{T^2} |c_n|^2 \quad (\because n^2 \geq 1 \text{ if } n \neq 0)$$

$$= \frac{4\pi^2}{T^2} \cdot T \sum_{n \neq 0} |c_n|^2$$

$$= \frac{4\pi^2}{T^2} \cdot \left(T \sum_{n \in \mathbb{Z}} |c_n|^2 \right) \quad (\because c_0 = 0)$$

$$= \frac{4\pi^2}{T^2} \|x\|^2$$

Wirtinger's inequality:

$$\text{If } \frac{1}{T} \int_0^T x(t) dt = 0$$

$$\text{Then } \|x'\|^2 \geq \frac{4\pi^2}{T^2} \|x\|^2$$